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Additive material integrated heater deposited or embedded within an ice detector

Abstract

A strut of a magnetostrictive oscillator includes a strut body which includes an airfoil with a first end, a second end, a leading edge, a trailing edge, a first side, and a second side. The strut further includes a heater element within the first side and second side, wherein the heater element connects from the first side to the second side. The strut further includes an electrically insulative layer between the heater element and the strut body. The heater element and the electrically insulative layer are integral with the strut body.

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Background/Summary

BACKGROUND

(1) The present disclosure relates to aircraft sensors, and in particular, to magnetostrictive oscillating (MSO) ice detector sensors (IDS).

(2) Aircraft sensors are important to proper operation of airplanes. Among these aircraft sensors are MSO IDS sensors which collect and detect liquid ice (supercooled water droplets) during flight. MSO IDS sensors can further detect the rate of ice accretion on an aircraft. Accurate information from these sensors is important to proper operation of the aircraft. During operation, these sensors accumulate ice on a detector probe and strut. To remove the ice on the detector probe and strut and reset the sensor, heaters heat the detector probe and strut to melt the ice off the detector probe and strut. Melting the ice off the detector probe and strut can be slow and energy intensive. Therefore, solutions to reduce power consumption and increase melting speed are desired.

SUMMARY

(3) In one embodiment, a strut of a magnetostrictive oscillating ice detector sensor includes a strut body with an airfoil. The airfoil includes a first end opposite a second end, and a leading edge extending from the first end to the second end. A trailing edge extends from the first end to the second end and is aft of the leading edge. The airfoil further includes a first side extending from the leading edge to the trailing edge and extending from the first end to the second end. A second side of the airfoil extends from the leading edge to the trailing edge and extends from the first end to the second end. The strut further includes a heater element within the first side and the second side, wherein the heater element connects from the first side to the second side. Strut further includes an electrically insulative layer between the heater element and the strut body. The heater element and the electrically insulative layer are integral with the strut body.

(4) In another embodiment, a method of forming a strut of a magnetostrictive oscillating ice detector sensor includes depositing an outside layer. Depositing the outside layer includes

depositing a first layer of powder, where the first layer of powder is a first material. Depositing the outside layer further includes sintering the first layer of powder. The method further includes depositing an insulator layer which includes depositing a second layer of powder on the outside layer. The second layer of powder includes a first portion of the second layer of powder which is the first material, a second portion of the second layer of powder which is a second material, and a third portion of the second layer of powder which is the first material. The second portion is between the first portion and the third portion. Depositing the insulator layer further includes sintering the second layer of powder. The method further includes depositing a heater element layer which includes depositing a third layer of powder on the insulating layer. The third layer of powder includes a first portion of the third layer of powder which is the first material, a second portion of the third layer of powder which is the second material, and a third portion of the third layer of powder which is a third material. The third layer of powder further includes a fourth portion of the third layer of powder which is the second material, and a fifth portion of the third layer of powder which is the first material. In the third layer of powder, the second portion is between the first portion and the third portion, and the fourth portion is between the third portion and the fifth portion. Depositing the heater element layer further includes sintering the third layer of powder. The method further includes depositing a second insulator layer above the heater element layer by repeating the insulator layer above the heater element layer, thereby forming a continuous conduit of the third material surrounded by the second material. The method further includes depositing a second outside layer above the second insulator layer by repeating the outside layer above the second insulator layer.

(5) In another embodiment, an ice detector sensor includes a mounting base, a support strut connected to the mounting base, a magnetostrictive oscillator probe head connected to the support strut opposite the mounting base, and a heater element within the support strut. The heater element is additively manufactured.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an embodiment of a magnetostrictive oscillating ice detector sensor.
- (2) FIG. 2A is a cross-sectional view of a probe of a magnetostrictive oscillating ice detector sensor where a heater element zigzags from a first end to a second end of the probe.
- (3) FIG. 2B is a cross-sectional view of a probe of a magnetostrictive oscillating ice detector sensor where a heater element spirals from a first end to a second end of the probe.
- (4) FIG. 2C is a cross-sectional view of the probe of the magnetostrictive oscillating ice detector sensor of FIG. 2A taken along line A-A.
- (5) FIG. 3A is a perspective view, with a cutout, of an embodiment of a probe of a magnetostrictive oscillating ice detector sensor with heater elements within a wall of the probe.
- (6) FIG. 3B is a cross-sectional view of the probe of the magnetostrictive oscillating ice detector sensor of FIG. 3A taken along line B-B and divided into additively manufacturable layers.
- (7) FIG. 4A is a perspective view, with a cutout, of an embodiment of a probe of a magnetostrictive oscillating ice detector sensor with heater elements inside a cavity of the probe.
- (8) FIG. 4B is a cross-sectional view of the probe of the magnetostrictive oscillating ice detector sensor of FIG. 4A taken along line C-C and divided into additively manufacturable layers.
- (9) FIG. 5A is a perspective view of a strut of a magnetostrictive oscillating ice detector sensor where a heater element zigzags from a first end to a second end of the strut.
- (10) FIG. 5B is a perspective view of a strut of a magnetostrictive oscillating ice detector sensor where a heater element spirals from a first end to a second end of the strut.

(11) FIG. 5C is a cross-sectional view of the strut of the magnetostrictive oscillating ice detector sensor of FIG. 5A taken along line D-D.

(12) FIG. 6A is a perspective view, with a cutout, of an embodiment of a strut of a magnetostrictive oscillating ice detector sensor with heater elements within a wall of the strut.

(13) FIG. 6B is a cross-sectional view of the strut of the magnetostrictive oscillating ice detector sensor of FIG. 6A taken along line E-E and divided into additively manufacturable layers.

(14) FIG. 7A is a perspective view, with a cutout, of an embodiment of a strut of a magnetostrictive oscillating ice detector sensor with heater elements inside a cavity of the strut.

(15) FIG. 7B is a cross-sectional view of the strut of the magnetostrictive oscillating ice detector sensor of FIG. 7A taken along line F-F and divided into additively manufacturable layers.

DETAILED DESCRIPTION

(16) FIG. 1 is a perspective view of an embodiment of magnetostrictive oscillating ice detector sensor **10**. Magnetostrictive oscillating ice detector sensor **10** comprises probe **12**, mounting base **22**, support strut **26**, and probe gasket **32**. Probe **12** comprises probe first end **14**, probe second end **16**, probe radially inner side **18** (not shown in FIG. 1), and probe radially outer side **20**. Mounting base **22** has attachment points **24** near an edge of mounting base **22**. Support strut **26** comprises strut first end **28** and strut second end **30**.

(17) Probe **12** of magnetostrictive oscillating ice detector sensor **10** is a magnetostrictive oscillator probe. Magnetostriction is the property of certain materials to expand and contract in response to a changing magnetic field. Magnetostrictive materials include ferromagnetic materials, nickel alloys, nickel-iron-chromium alloys, and NiSPAN 902. Magnetostrictive oscillating ice detector sensor **10** functions by expanding and contracting probe **12** under a variable magnetic field. As ice builds up on probe radially outer side **20**, the added mass of ice on probe **12** causes the frequency of probe **12** to decrease while under the same oscillating magnetic field. Therefore, the thickness of the ice on probe **12** can be approximated based on the decrease in frequency. Once the frequency has decreased beyond a set point, a heating element will be turned on to heat probe **12**, thereby melting the accumulated ice and resetting probe **12**.

(18) Probe **12** comprises probe first end **14** which is opposite probe second end **16**. Probe **12** can be formed of a cylindrical portion which extends from probe first end **14** towards probe second end **16**. The cylindrical portion can be capped by a hemispherical portion at the probe first end **14**. Hemispherical portion at probe first end **14** can reduce ice accumulation at probe first end **14**. Alternatively, probe **12** can be formed of a cylindrical portion which extends from probe first end **14** to probe second end **16**. Alternatively, probe **12** can be formed of an airfoil shape which extends from probe first end **14** to probe second end **16**. Probe **12** has probe radially inner side **18** (shown in FIGS. 2A-2C) and probe radially outer side **20**. Probe radially outer side **20** contacts an exterior environment and during operation will accumulate ice. Probe radially outer side **20** can be smooth. The probe radially outer side **20** can be polished to increase a smoothness of the surface. An increased smoothness of probe radially outer side **20** can reduce the time required to melt accumulated ice off probe radially outer side **20**. Alternatively, a surface treatment can be applied to probe **12** to increase a surface roughness of probe radially outer side **20**. An increased roughness of probe radially outer side **20** can increase an ice accumulation rate.

(19) Strut **26** has strut first end **28** opposite strut second end **30**. Probe second end **16** connects to strut **26** at strut second end **30**. The connection between probe second end **16** and strut second end **30** can be reinforced by probe gasket **32**. Probe gasket **32** reduces leakage between probe second end **16** and strut second end **30**. Specifically, probe gasket **32** can reduce the infiltration of water between a gap between strut second end **30** and probe second end **16**. Probe gasket **32** can be formed of rubber, plastic, metal, or other materials known to those of skill in the art to seal a gap. Probe gasket **32** cannot halt movement of probe **12** as halting movement of probe **12** removes the ability of probe **12** to oscillate under an alternating magnetic field as described above. Strut **26** connects to mounting base **22** at strut first end **28**. Strut **26** can be shaped as an airfoil.

Alternatively, strut **26** can be cylindrically shaped. Alternatively, strut **26** can be oval shaped. Mounting base **22** can be affixed to a larger system via attachment points **24**. Attachment points **24** can be used for reversible attachment mechanisms such as screws and bolts. Alternatively, attachment points **23** can be used for irreversible attachment mechanisms such as rivets, welding, or brazing. The larger system can be an aircraft, such as an airplane. Specifically, the mounting base can be affixed to a fuselage near a nose of an airplane.

(20) FIGS. 2A-2C disclose alternative arrangements for heater element **38** in probe **12** and will be discussed together. FIG. 2A is a cross-sectional view of probe **12** of magnetostrictive oscillating ice detector sensor **10** where heater element **38** zigzags from probe first end **14** to probe second end **16**. FIG. 2B is a cross-sectional view of probe **12** of magnetostrictive oscillating ice detector sensor **10** where heater element **38** spirals from first end **14** to second end **16**. FIG. 2C is a cross-sectional view of probe **12** of magnetostrictive oscillating ice detector sensor **10** of FIG. 2A taken along line A-A. Probe **12** includes probe first end **14**, probe second end **16**, probe radially inner side **18**, and probe radially outer side **20**.

(21) As best shown in FIG. 2A, heater element zig-zag pattern **34** comprises heater element **38** and insulative element **40** zigzagging from probe first end **14** to probe second end **16**. Starting from probe second end **16** heater element zig-zag pattern **34** comprises heater element **38** wrapping circumferentially a first circumferential distance in probe **12**, then heater element **38** extends towards probe first end **14** a first linear distance. Heater element **38** will then wrap circumferentially a second circumferential distance and then extend for a second linear distance. The wrapping and extending steps repeat until heater element **38** reaches probe first end **14**. The first circumferential and the second circumferential distances that heater element **38** circumferentially wraps can be less than a full circumference of probe **12**. Alternatively, the first circumferential and the second circumferential distances that the probe circumferentially wraps can be greater than a circumference of probe **12**. The first circumferential distance and the second circumferential distance can be the same. Alternatively, the first circumferential distance and the second circumferential distance can be different. A sum of the first linear distance and the second linear distance cannot be greater than a distance from probe first end **14** to probe second end **16**. A sum of all linear distances over which heater element **38** extends can be equal to a distance between probe first end **14** and probe second end **16**. Heater element **38** can run along radially inner side **18** of probe **12**. Alternatively, heater element **38** can run within a wall of probe **12**. Heater element **38** can be surrounded by insulative element **40**. Heater element zig-zag pattern **34** can be optimized so that the linear distances between circumferential wrappings varies over the distance from probe first end **14** to probe second end **16**. The optimization enables portions of probe **12** to be heated to a greater degree than other portions of probe **12**.

(22) As best shown in FIG. 2B, heater element spiral pattern **36** comprises heater element **38** and insulative element **40** spiraling from probe first end **14** to probe second end **16** at Slope α . Starting from probe second end **16** heater element spiral pattern **36** comprises heater element **38** wrapping circumferentially in probe **12** at slope α towards first end **14**. Slope α is an angle between a plane perpendicular to a central axis of probe **12** and a line parallel to heater element **38** and insulative element **40**. Slope α can be a grade of greater than 1%, greater than 3%, or greater than 5%. Slope α can be dependent on a resistance of heater element **38**. Slope α can be dependent on the wattage required to heat probe **12**. Heater element **38** can run along radially inner side **18** of probe **12**. Alternatively, heater element **38** can run within a wall of probe **12**.

(23) As best shown in FIG. 2C, heater element **38** has thickness d . Thickness d can be varied throughout the length that heater element **38** runs. Specifically, heater element **38** can be made thicker to reduce heat production in specific areas. Alternatively, heater element **38** can be made thinner to increase heat production in other areas. Probe **12** has circumference A . As discussed above with respect to FIG. 2A, heater element zig-zag pattern **34** can circumferentially wrap all of circumference A as shown in FIG. 2C. Alternatively, heater element **38** can wrap around probe **12** a

fraction of circumference A.

(24) Both heater element zig-zag pattern **34** and heater element spiral pattern **36** enable heating element **38** to distribute heat produced by heating element **38** throughout fore, aft, and in-between locations of probe **12**. Alternatively, to the patterns shown in FIGS. 2A and 2B, a combination of heater element zig-zag pattern **34** and heater element spiral pattern **36** can be employed.

Specifically, heater element **38** can employ heater element zig-zag pattern **34** for a portion and transition to heater element spiral pattern **36** for a second portion. Alternatively, heater element zig-zag pattern **34** and heater element spiral pattern **36** can be combined where the circumferential wraps of zig-zag pattern **34** have slope α of heater element spiral pattern **36**.

(25) FIGS. 3A and 3B discuss an embodiment where heater element **38** is embedded in probe wall **42** thereby forming probe wall heater **44**. FIG. 3A is a perspective view, with a cutout, of an embodiment of probe **12** of magnetostrictive oscillating ice detector sensor **10** with heater elements **38** within probe wall **42**. FIG. 3B is a cross-sectional view of probe **12** of magnetostrictive oscillating ice detector sensor **10** of FIG. 3A taken along line B-B and divided into additively manufacturable layers **50**. Probe **12** includes probe first end **14**, probe second end **16**, probe radially inner side **18**, probe radially outer side **20**, and probe wall **42**. Within probe wall **42** is probe wall heater **44** formed from heater element **38** surrounded by insulative element **40**. Probe wall heater **44** can be formed by many additively manufacturable layers **50**.

(26) As best shown in FIG. 3A, probe **12** has probe first end **14** and probe second end **16**. Probe **12** has radially inner side **18** and radially outer side **20**. Between radially inner side **18** and radially outer side **20** and extending from probe first end **14** to probe second end **16** is probe wall **42**. Within probe wall **42** is formed heater element **38** surrounded by insulative element **40**. By forming heater element **38** in probe wall **42**, probe wall heater **44** is formed. Heater element **38** can be formed inside probe wall **42** by additively manufacturing probe **12**. Insulative element **40** can be formed around heater element **38** when additively manufacturing probe **12**. As discussed above with respect to FIG. 1, probe can be any shape known to those of skill in the art as being functional probes for magnetostrictive oscillating ice detector sensors **10**.

(27) As best shown in FIG. 3B, probe wall heater **44** can be formed by a plurality of additively manufacturable layers **50**. Each additively manufacturable layer **50** can have a different composition. Specifically, each additively manufacturable layer **50** can have a percentage which is a magnetostrictive material, a percentage which is an insulative material, and a percentage which is a heater material. The plurality of additively manufacturable layers **50** can be broken into three main types of layers. The three main types of layers are an outside layer, an insulator layer, and a heater element layer.

(28) The outside layer OL comprises a first layer of powder which is a first material. The first material can be a magnetostrictive material such as a ferromagnetic metal, a ferromagnetic alloy, a nickel alloy, nickel-iron-chromium alloy, NiSPAN 920, and combinations thereof. Once the powder has been laid, the powder is then sintered by a high-powered laser. The insulator layer IL comprises depositing a second layer of powder onto the outside layer OL. The insulator layer IL has a first portion and a third portion which are formed of the first material. The insulator layer has a second portion between the first and the third portions which is formed of a second material. The second material can be an insulative material. The insulative material can be ceramic, plastic, rubber, and combinations thereof. The insulative material can be any insulative material known to those of skill in the art as having a sufficient dielectric with-standing to reduce a current therethrough. Once the powder has been laid, the powder is then sintered by a high-powered laser. The heater element layer HL comprises depositing a third layer (or more) of powder onto the insulator layer IL. Each heater element layer HL comprises a first and a fifth portion which are formed of the first material, a second and fourth portion which are formed of the second material, and a third portion which is formed of a third material. The second portion is between the first and third portions while the fourth portion is between the third and fifth portions. The third material can be a heater element

material. The heater element material can be a nichrome alloy, metal alloys, ceramic materials, ceramic metals, and combinations thereof. The heater element material can be any material known to those of skill in the art as producing heat when resisting an electric current passed through the material. Once the powder has been laid, the powder is then sintered by a high-powered laser. The high-powered laser can be a 200-watt laser. The high-powered laser can be a Yb-fiber optic laser. Other power levels or laser types known to those of skill in the art as being able to sinter metal powder, ceramic powder, or plastic powders can be used.

(29) A second insulator layer IL can be placed above the heater element layer(s) HL by repeating the insulator layer IL steps detailed above. By placing the second insulator layer IL, a continuous conduit of the third material is surrounded by a continuous layer of the second material. By having a continuous conduit of the second material around the third material, heater element **38** is electrically insulated by insulative element **40** from probe **12**. A second outside layer OL can be placed above the second insulator layer IL by repeating the outside layer OL steps detailed above. The second outside layer OL enables probe radially outer side **20** and probe radially inner side **18** to be formed solely of the first material. As such, none of the insulator material or the heater element material will be exposed to an outside atmosphere in the embodiment of FIGS. **3A** and **3B**. Probe wall heater **44** can be formed in heater element zig-zag pattern **34** or heater element spiral pattern **36**. As discussed above with respect to FIGS. **2A-2C**, other patterns or combinations of patterns can be used.

(30) FIGS. **4A** and **4B** discuss an embodiment where heater element **38** is formed in probe cavity **46** on probe radially inner side **18** thereby forming probe cavity heater **48**. FIG. **4A** is a perspective view, with a cutout, of an embodiment of probe **12** of magnetostrictive oscillating ice detector sensor **10** with heater elements **38** inside a cavity of probe **12**. FIG. **4B** is a cross-sectional view of probe **12** of magnetostrictive oscillating ice detector sensor **10** of FIG. **4A** taken along line C-C and showing probe cavity heater **48** divided into additively manufacturable layers **50**. Probe **12** includes probe first end **14**, probe second end **16**, probe radially inner side **18**, probe radially outer side **20**, and probe cavity **46**. Within probe cavity **46** is probe cavity heater **48**. Probe cavity heater **48** is formed from heater element **38** which is formed on insulative element **40**. Insulative element **40** is formed on probe radially inner side **18**. Probe cavity heater **48** can be formed by many additively manufacturable layers **50**.

(31) As best shown in FIG. **4A**, probe **12** has probe first end **14** and probe second end **16**. Probe **12** has radially inner side **18** and radially outer side **20**. Within probe radially inner side **18** is probe cavity **46**. Within probe cavity **46** is formed heater element **38**. Between heater element **38** and radially inner side **18** is insulative element **40**. By forming heater element **38** within probe cavity **46**, probe cavity heater **48** is formed. Heater element **38** can be formed inside probe cavity **46** by additively manufacturing heater element **38** and probe **12**. Insulative element **40** can be formed between heater element **38** and radially inner side **18** when additively manufacturing probe **12**. Alternatively, heater element **38** can be additively manufactured into probe cavity **46** when probe **12** is formed via traditional manufacturing methods. As discussed above with respect to FIG. **1**, probe **12** can be any shape known to those of skill in the art as being functional probes **12** for magnetostrictive oscillating ice detector sensors **10**.

(32) As best shown in FIG. **4B**, heater element **38** and insulative element **40** formed in probe cavity **46** can be described by a plurality of additively manufacturable layers **50**. Each additively manufacturable layer **50** can have a different composition. The plurality of additively manufacturable layers **50** can be broken into two main types of layers. The two main types of layers are an insulator layer, and a heater element layer.

(33) The insulative layer is formed by placing a powder of a first material onto probe radially inner side **18**. The first material can be ceramic, plastic, rubber, and combinations thereof. Alternatively, the first material can be any material known to those of skill in the art as having a sufficient dielectric with-standing to reduce a current therethrough. Once the powder has been laid, the

powder is then sintered by a high-powered laser. The heater element layer is formed by placing a powder of a second material onto the insulative layer. The second material can be a nichrome alloy, metal alloys, ceramic materials, ceramic metals, and combinations thereof. The second material can be any material known to those of skill in the art as producing heat when resisting an electric current passed through the material. Once the powder has been laid, the powder is then sintered by a high-powered laser. The high-powered laser can be a 200-watt laser. The high-powered laser can be a Yb-fiber optic laser. Other power levels or laser types known to those of skill in the art as being able to sinter metal powder, ceramic powder, or plastic powders can be used.

(34) Alternatively, the insulative layer can be formed onto probe radially inner side **18** via a spray deposition process. The spray deposition process includes forcing a high velocity stream of inert gas through a nozzle tip. Near the nozzle tip a stream of molten material is introduced. The high velocity stream carries the molten material from the nozzle tip to a deposition point on the radially inner side **18**. The molten material rapidly solidifies as it travels from the nozzle tip to a deposition point on the radially inner side **18**. When forming the insulative layer, the molten material can be a ceramic, plastic, rubber, and combinations thereof. Alternatively, the insulative material can be any insulative material known to those of skill in the art as having a sufficient dielectric with-standing to reduce a current therethrough. After formation of the insulative layer, a heater element layer is formed onto the insulative layer via the spray deposition process. When forming the heater element layer, the molten material can be a nichrome alloy, metal alloys, ceramic materials, ceramic metals, and combinations thereof. The heater element material can be any material known to those of skill in the art as producing heat when resisting an electric current passed through the material. The inert gas can be any gas or combination of gas which does not adversely react with the molten material.

(35) FIGS. 5A-5C disclose alternative arrangements for heater element **38** in strut **26** and will be discussed together. FIG. 5A is a perspective view of strut **26** of magnetostrictive oscillating ice detector sensor **10** where heater element **38** zigzags from strut first end **28** to strut second end **30**. FIG. 5B is a perspective view of strut **26** of magnetostrictive oscillating ice detector sensor **10** where heater element **38** spirals from strut first end **28** to strut second end **30**. FIG. 5C is a cross-sectional view of strut **26** of magnetostrictive oscillating ice detector sensor **10** of FIG. taken along line D-D. Strut **26** includes strut first end **28**, strut second end **30**, strut leading edge **52**, strut trailing edge **54**, strut first side **56**, and strut second side **58**.

(36) As best shown in FIG. 5A, heater element zig-zag pattern **34** comprises heater element **38** and insulative element **40** zigzagging from strut first end **28** to strut second end **30**. Starting from strut second end **28** heater element zig-zag pattern **34** comprises heater element **38** wrapping circumferentially a first circumferential distance in strut **26** toward trailing edge **54** of strut **26**, then heater element **38** extends towards probe second end **30** a first linear distance. Heater element **38** will then wrap circumferentially a second circumferential distance toward leading edge **52** of strut **26**, and then extend for a second linear distance. The wrapping and extending steps repeat until heater element **38** reaches probe second end **30**. The first circumferential and the second circumferential distances that heater element **38** circumferentially wraps can be less than a circumference of strut **26**. Alternatively, the first circumferential and the second circumferential distances that the probe circumferentially wraps can be greater than a circumference of strut **26**. The first circumferential distance and the second circumferential distance can be the same. Alternatively, the first circumferential distance and the second circumferential distance can be different. A sum of the first linear distance and the second linear distance cannot be greater than a distance from strut first end **28** to strut second end **30**. A sum of all linear distances over which heater element **38** extends must be equal to a distance between strut first end **28** and strut second end **30**. Heater element **38** can run along an inside surface of strut **26**. Alternatively, heater element **38** can run within a wall of strut **26**. Heater element **38** can be surrounded by insulative element **40**. Heater element zig-zag pattern **34** can be optimized so that the linear distances between circumferential wrappings varies over the distance from strut first end **28** to strut second end **30**.

The optimization enables portions of probe **12** to be heated to a greater degree than other portions of probe **12**.

(37) As best shown in FIG. 5B, heater element spiral pattern **36** comprises heater element **38** and insulative element **40** spiraling from strut first end **28** to strut second end **30** at slope β . Starting from strut first end **28** heater element spiral pattern **36** comprises heater element **38** wrapping circumferentially in strut **26** at slope β towards strut second end **30**. Slope β is an angle between a plane perpendicular to a central axis of strut **26** and a line parallel to heater element **38** and insulative element **40**. Slope β can be a grade of greater than 1%, greater than 3%, or greater than 5%. Slope β can be dependent on a resistance of heater element **38**. Slope β can be dependent on the wattage required to heat strut **26**. Heater element **38** can run along an inside surface of strut **26**. Alternatively, heater element **38** can run within a wall of strut **26**.

(38) As best shown in FIG. 5C, heater element **38** has thickness d . Thickness d can be varied throughout the length that heater element **38** runs. Specifically, heater element **38** can be made thicker to reduce heat production in specific areas. Alternatively, heater element **38** can be made thinner to increase heat production in other areas. Strut **26** has leading edge **52** and trailing edge **54**. Between leading edge **52** and trailing edge **54** is an airfoil distance, also referred to as chord length. Strut **26** also has first side **56** and second side **58**. Between first side **56** and second side **58** is an airfoil width. As discussed above with reference to FIG. 1, strut **26** can be shaped as an airfoil. If strut **26** is shaped as an airfoil, then the airfoil distance will be greater than the airfoil width. Alternatively, strut **26** can be shaped as a cylinder. If strut **26** is shaped as a cylinder, then the airfoil width will be equal to the airfoil length. As shown in FIG. 5C, heater element **38** can wrap around a whole circumference of strut **26**. Alternatively, heater element **38** can wrap around a fraction of the circumference of strut **26**.

(39) Both heater element zig-zag pattern **34** and heater element spiral pattern **36** enable heating element **38** to distribute heat produced throughout fore, aft, and in-between locations of strut **26**. Alternatively, to the patterns shown in FIGS. 5A and 5B, a combination of heater element zig-zag pattern **34** and heater element spiral pattern **36** can be employed. Specifically, heater element **38** can employ heater element zig-zag pattern **34** for a portion and transition to heater element spiral pattern **36** for a second portion. Alternatively, heater element zig-zag pattern **34** and heater element spiral pattern **36** can be combined where the circumferential wraps of zig-zag pattern **34** have slope β of heater element spiral pattern **36**.

(40) FIGS. 6A and 6B discuss an embodiment where heater element **38** is embedded in strut wall **60** thereby forming strut wall heater **62**. FIG. 6A is a perspective view, with a cutout, of an embodiment of strut **26** of magnetostrictive oscillating ice detector sensor **10** with heater elements **38** within strut wall **60**. FIG. 6B is a cross-sectional view of strut **26** of magnetostrictive oscillating ice detector sensor **10** of FIG. 6A taken along line B-B and divided into additively manufacturable layers **50**. Strut **26** includes strut first end **28**, strut second end **30**, strut leading edge **52**, strut trailing edge **54**, strut first side **56**, strut second side **58**, and strut wall **60**. Within strut wall **60** is strut wall heater **62** formed from heater element **38** surrounded by insulative element **40**. Strut wall heater **62** can be formed by many additively manufacturable layers **50**.

(41) As best shown in FIG. 6A, strut **26** has strut first end **28** and strut second end **30**. Strut **26** has leading edge **52** and trailing edge **54**. Strut has first side **56** (not shown in FIG. 7A) and second side **58**. Extending from strut first end **28** to strut second end **30** is strut wall **60** with a wall thickness. Within strut wall **60** is formed heater element **38** surrounded by insulative element **40**. By forming heater element **38** in strut wall **60**, strut wall heater **62** is formed. Heater element **38** can be formed inside strut wall **60** by additively manufacturing strut **26**. Insulative element **40** can be formed around heater element **38** when additively manufacturing strut **26**. As discussed above with respect to FIG. 1, strut **26** can be any shape known to those of skill in the art as being functional struts **26** for magnetostrictive oscillating ice detector sensors **10**.

(42) As best shown in FIG. 6B, strut wall heater **62** can be formed by a plurality of additively

manufacturable layers **50**. Each additively manufacturable layer **50** can have a different composition. Specifically, each additively manufacturable layer **50** can have a percentage which is a base material, a percentage which is an insulative material, and a percentage which is a heater material. The plurality of additively manufacturable layers **50** can be broken into three main types of layers. The three main types of layers are an outside layer OL, an insulator layer IL, and a heater element layer HL.

(43) The outside layer OL comprises a first layer of powder which is a first material. The first material can be any aerospace material suitable to experience the temperatures and pressures of flight conditions. Materials such as aluminum, aluminum alloy, titanium, composites, and combinations thereof can be suitable aerospace materials. The first material can be any material known to those of skill in the art as capable of conducting heat through the material. Once the powder has been laid, the powder is then sintered by a high-powered laser. The insulator layer IL comprises depositing a second layer of powder onto the outside layer OL. The insulator layer IL has a first portion and a third portion which are formed of the first material. The insulator layer has a second portion between the first and the third portions which is formed of a second material. The second material can be an insulative material. The insulative material can be ceramic, plastic, rubber, and combinations thereof. The insulative material can be any insulative material known to those of skill in the art as having a sufficient dielectric with-standing to reduce a current therethrough. Once the powder has been laid, the powder is then sintered by a high-powered laser. The heater element layer HL comprises depositing a third layer of powder onto the insulator layer IL. The heater element layer HL comprises a first and a fifth portion which are formed of the first material, a second and fourth portion which are formed of the second material, and a third portion which is formed of a third material. The second portion is between the first and third portions while the fourth portion is between the third and fifth portions. The third material can be a heater element material. The heater element material can be a nichrome alloy, metal alloys, ceramic materials, ceramic metals, and combinations thereof. The heater element material can be any material known to those of skill in the art as producing heat when resisting an electric current passed through the material. Once the powder has been laid, the powder is then sintered by a high-powered laser. The high-powered laser can be a 200-watt laser. The high-powered laser can be a Yb-fiber optic laser. Other laser power levels or laser types known to those of skill in the art as being able to sinter metal powder, ceramic powder, or plastic powders can be used.

(44) A second insulator layer IL can be placed above the heater element layer HL by repeating the insulator layer IL steps detailed above. By placing the second insulator layer IL, a continuous conduit of the third material is surrounded by a continuous layer of the second material. By having a continuous conduit of the second material around the third material, heater element **38** is electrically insulated by insulative element **40** from strut **26**. A second outside layer OL can be placed above the second insulator layer IL by repeating the outside layer OL steps detailed above. The first and second outside layers OL enable an inside surface and an outside surface of strut **26** to be formed solely of the first material. As such, none of the insulator material or the heater element material will be exposed to an outside atmosphere in the embodiment of FIGS. **6A** and **6B**. Strut wall heater **62** can be formed in heater element zig-zag pattern **34** or heater element spiral pattern **36**. As discussed above with respect to FIGS. **5A-5C**, other patterns or combinations of patterns can be used.

(45) FIGS. **7A** and **7B** discuss an embodiment where heater element **38** is formed in strut cavity **64** on an inner side of strut **26** thereby forming strut cavity heater **66**. FIG. **7A** is a perspective view, with a cutout, of an embodiment of strut **26** of magnetostrictive oscillating ice detector sensor **10** with heater elements **38** inside strut cavity **64**. FIG. **7B** is a cross-sectional view of strut **26** of magnetostrictive oscillating ice detector sensor **10** of FIG. **7A** taken along line C-C and divided into additively manufacturable layers **50**. Strut **26** includes strut first end **28**, strut second end **30**, strut leading edge **52**, strut trailing edge **54**, strut first side **56**, strut second side **58**, and strut cavity

64. Within strut cavity **64** is strut cavity heater **66**. Strut cavity heater **66** is formed from heater element **38** which is formed on insulative element **40**. Insulative element **40** is formed on inside faces of strut first side **56** and strut second side **58**. Strut cavity heater **66** can be formed by many additively manufacturable layers **50**.

(46) As best shown in FIG. 7A, strut **26** has strut first end **28** and strut second end **30**. Strut **26** has leading edge **52** and trailing edge **54**. Strut has strut first side **56** (not shown in FIG. 7A) and strut second side **58**. Strut cavity **64** is formed between strut first end **28**, strut second end strut leading edge **52**, strut trailing edge **54**, strut first side **56**, and strut second side **58**. Within strut cavity **64**, formed touching an inside face of strut first side **56** and strut second side **58**, is heater element **38**. Between heater element **38** and the inside face of strut first side **56** and strut second side **58** is insulative element **40**. By forming heater element **38** within strut cavity **64**, strut cavity heater **66** is formed. Heater element **38** can be formed inside strut cavity **64** by additively manufacturing heater element **38** and strut **26**. Insulative element **40** can be formed between heater element **38** and the inside face of strut first side **56** and strut second side **58** when additively manufacturing strut **26**. Alternatively, heater element **38** can be additively manufactured into strut cavity **64** after strut **26** is formed via traditional manufacturing methods. As discussed above with respect to FIG. 1, strut **26** can be any shape known to those of skill in the art as being functional as a strut for an aircraft.

(47) As best shown in FIG. 7B, heater element **38** and insulative element **40** formed in strut cavity **64** can be formed by a plurality of additively manufacturable layers **50**. Each additively manufacturable layer **50** can have a different composition. The plurality of additively manufacturable layers **50** can be classified into two main types of layers. The two main types of layers are an insulator layer, and a heater element layer.

(48) The insulative layer is formed by placing a powder of a first material onto the inside face of strut first side **56** and strut second side **58**. The first material can be ceramic, plastic, rubber, and combinations thereof. Alternatively, the first material can be any material known to those of skill in the art as having a sufficient dielectric with-standing to reduce a current therethrough. Once the powder has been laid, the powder is then sintered by a high-powered laser. The heater element layer is formed by placing a powder of a second material onto the insulative layer. The second material can be a nichrome alloy, metal alloys, ceramic materials, ceramic metals, and combinations thereof. The second material can be any material known to those of skill in the art as producing heat when resisting an electric current passed through the material. Once the powder has been laid, the powder is then sintered by a high-powered laser. The high-powered laser can be a 200-watt laser. The high-powered laser can be a Yb-fiber optic laser. Other power levels or laser types known to those of skill in the art as being able to sinter metal powder, ceramic powder, or plastic powders can be used. Strut cavity heater **66** can be formed in heater element zig-zag pattern **34** or heater element spiral pattern **36**. As discussed above with respect to FIGS. 5A-5C, other patterns or combinations of patterns can be used.

(49) Alternatively, the insulative layer can be formed onto the inside face of strut first side **56** and strut second side **58** via a spray deposition process. The spray deposition process includes forcing a high velocity stream of inert gas through a nozzle tip. Near the nozzle tip a stream of molten material is introduced. The high velocity stream carries the molten material from the nozzle tip to a deposition point on the inside face of strut first side **56** and strut second side **58**. The molten material rapidly solidifies as it travels from the nozzle tip to a deposition point on the inside face of strut first side **56** and strut second side **58**. When forming the insulative layer, the molten material can be a ceramic, plastic, rubber, and combinations thereof. Alternatively, the insulative material can be any insulative material known to those of skill in the art as having a sufficient dielectric with-standing to reduce a current therethrough. After formation of the insulative layer, a heater element layer is formed onto the insulative layer via the spray deposition process. When forming the heater element layer, the molten material can be a nichrome alloy, metal alloys, ceramic materials, ceramic metals, and combinations thereof. The heater element material can be any

material known to those of skill in the art as producing heat when resisting an electric current passed through the material. The inert gas can be any gas or combination of gas which does not adversely react with the molten material.

Discussion of Possible Embodiments

(50) The following are non-exclusive descriptions of possible embodiments of the present invention.

(51) A strut of a magnetostrictive oscillating ice detector sensor includes a strut body including an airfoil with a first end opposite a second end, a leading edge extending from the first end to the second end, a trailing edge extending from the first end to the second end and aft of the leading edge, a first side extending from the leading edge to the trailing edge and extending from the first end to the second end, and a second side extending from the leading edge to the trailing edge and extending from the first end to the second end. The strut further includes a heater element within the first side and the second side, the heater element connects from the first side to the second side, an electrically insulative layer between the heater element and the strut body, and the heater element and the electrically insulative layer are integral with the strut body.

(52) The strut of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(53) The heater element, and the insulative layer are additively manufactured.

(54) The heater element is in a wall of the first side.

(55) The heater element is on a radially inward face of the first side and the second side.

(56) The strut body is formed of an aluminum alloy.

(57) The heater element spirals from the first end to the second end of the strut body.

(58) The heater element zigzags from the first end to the second end of the strut body.

(59) A method of forming a strut of a magnetostrictive oscillating ice detector sensor, the method includes depositing an outside layer, including depositing a first layer of powder, wherein the first layer of powder is a first material, sintering the first layer of powder. The method further includes depositing an insulator layer including depositing a second layer of powder on the outside layer, a first portion of the second layer of powder is the first material, a second portion of the second layer of powder is a second material, a third portion of the second layer of powder is the first material, and the second portion is between the first portion and the third portion, sintering the second layer of powder. The method further includes depositing a heater element layer including depositing a third layer of powder on the insulating layer, a first portion of the third layer of powder is the first material, a second portion of the third layer of powder is the second material, a third portion of the third layer of powder is a third material, a fourth portion of the third layer of powder is the second material, and a fifth portion of the third layer of powder is the first material. The second portion is between the first portion and the third portion, the fourth portion is between the third portion and the fifth portion, and sintering the third layer of powder. The method further includes depositing a second insulator layer above the heater element layer by repeating the insulator layer above the heater element layer, thereby forming a continuous conduit of the third material surrounded by the second material and depositing a second outside layer above the second insulator layer by repeating the outside layer above the second insulator layer.

(60) The method of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(61) The first material is an aluminum alloy.

(62) The second material is an electrically insulative material.

(63) The third material is a resistive heating material.

(64) The first material forms a body of the strut.

(65) The continuous conduit of the third material forms a heater element.

(66) The heater element runs from a first end of the strut to a second end of the strut.

- (67) An ice detector sensor includes a mounting base, a support strut connected to the mounting base, a magnetostrictive oscillator probe head connected to the support strut opposite the mounting base, and a heater element within the support strut, wherein the heater element is additively manufactured.
- (68) The ice detector sensor of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:
- (69) The mounting base connects to an aircraft.
- (70) The support strut is airfoil shaped.
- (71) The support strut and the heater element are additively manufactured together.
- (72) The heater is within a wall of the support strut.
- (73) The heater element is inside a cavity of the support strut.

Claims

1. A strut of a magnetostrictive oscillating ice detector sensor comprising: a strut body comprising: an airfoil with a first end opposite a second end, a leading edge extending from the first end to the second end, a trailing edge extending from the first end to the second end and aftward of the leading edge, a first side extending from the leading edge to the trailing edge and extending from the first end to the second end, and a second side extending from the leading edge to the trailing edge and extending from the first end to the second end; and a cavity formed between a first inside face of the first side and a second inside face of the second side, between the leading edge and the trailing edge, and between the first end and the second end; a heater element within the first side and the second side, wherein the heater element is on the first inside face of the first side and/or on a second inside face of the second side; and an electrically insulative layer between the heater element and the strut body; wherein the electrically insulative layer and the heater element form a heater element layer that spans a length of the cavity.
2. The strut of claim 1, wherein the strut body, the heater element, and the insulative layer are additively manufactured.
3. The strut of claim 1, wherein the heater element is in a wall on the first inside face of the first side.
4. The strut of claim 1, wherein the strut body is formed of an aluminum alloy.
5. The strut of claim 1, wherein the heater element spirals from the first end to the second end of the strut body.
6. The strut of claim 1, wherein the heater element zigzags from the first end to the second end of the strut body.
7. A strut of a magnetostrictive oscillating ice detector sensor comprising: a wall comprising: a first side comprising a first inside face and a first outside face; and a second side comprising a second inside face and a second outside face; a cavity formed between the first inside face of the first side and the second inside face of the second side; a first insulative layer on the first inside face of the first side, wherein the first insulative layer is additively formed onto the first inside face of the first side; and a first conductive layer on the first insulative layer, wherein the first conductive layer is additively formed onto the first insulative layer, and wherein the first insulative layer and the first conductive layer form a heater element layer that spans a length of the cavity.
8. The strut of the magnetostrictive oscillating ice detector sensor of claim 7, and further comprising: a second insulative layer on the second inside face of the second side, wherein the second insulative layer is additively formed onto the second inside face of the second side; and a second conductive layer on the second insulative layer, wherein the second conductive layer is additively formed onto the second insulative layer, and wherein the first conductive layer is electrically connected to the second conductive layer.

9. The strut of the magnetostrictive oscillating ice detector sensor of claim 8, wherein the first insulative layer is connected to the second insulative layer.
10. The strut of the magnetostrictive oscillating ice detector sensor of claim 9, wherein the first insulative layer and the second insulative layer are formed via a spray deposition process.
11. The strut of the magnetostrictive oscillating ice detector sensor of claim 9, wherein the first conductive layer and the second conductive layer are formed via a spray deposition process.
12. The strut of the magnetostrictive oscillating ice detector sensor of claim 7, wherein the first insulative layer is electrically insulative but thermally conductive.
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